# Brendan Ling

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## Education

**University of Manchester** — Manchester, UK
*BSc Computer Science with Honours* — *Sep. 2024 – Jun. 2027*

- First Class expected, **ranked 1st in a cohort of approx. 250** for second year with an **average of 89.6%.**
- Awarded the **Best Overall Performance – 1st Prize** in second year, and the **Netcraft Prize** in first year awarded to the top 10 students (placed 5th).
- Recipient of the **DSTA Overseas Scholarship**, a full-ride scholarship from Singapore's Defence Science and Technology Agency.
- Relevant Coursework: **Machine Learning (98%), Introduction to Artificial Intelligence (90%), Distributed Systems (90%), Algorithms & Data Structures (87%).**

## Research

**Backdoor Attacks on Federated Learning Defences** — Sep. 2026 – Present
*University of Manchester* | *Supervisors: Dr. Zhipeng Wang, Dr. Mustafa Mustafa* — *Manchester, UK*

- Developing a backdoor attack against federated learning defences, using **FilterFL (CCS 2025)** as the primary case study.
- Designed a full-image checkerboard trigger that evades FilterFL as well as **FLTrust (NDSS 2021)** while preserving clean accuracy; a single accepted malicious update is enough to entrench the backdoor, which persists under benign training.
- Showed that confidence values assigned by FilterFL's detector primarily measure a client's **class skew rather than backdoor presence**, causing it to flag honest clients more reliably than successful attackers.
- Built an independent reimplementation and ran controlled ablations on HPC (**CSF3**).

**Artificial Intelligence Research Intern** — Jun. 2025 – Aug. 2025
*Defence Science and Technology Agency* — *Singapore, SG*

- Built an LLM-agent social media simulation on the open-source **OASIS** framework, modelling **hundreds of Reddit users** to study misinformation propagation.
- Designed **agent personas** and the **platform's mock recommendation algorithm**; tracked per-post engagement to analyse what drives virality.
- Identified a fidelity gap between LLM-agent and human behaviour: overall **accuracy reached 82% but was driven by true negatives**, as agents favoured emotionally charged content real users did not amplify.

## Professional Experience

**Artificial Intelligence Development Intern** — Jun. 2026 – Aug. 2026
*Defence Science and Technology Agency* — *Singapore, SG*

- Architected a **proof-of-concept multi-agent logistics control tower**: three specialised LLM agents orchestrate **17 tools** to turn natural-language operator requests into structured routing and dispatch actions.
- Separated probabilistic agent reasoning from deterministic execution via a stateful model, with **schema-validation guardrails** and a **rules-engine fallback**.
- Implemented fleet optimisation with **OR-Tools** (CVRP/VRPTW) alongside constrained **NetworkX** shortest-path routing for single-vehicle requests.
- Evaluated the agents with **DeepEval** across **30 test scenarios**, achieving **90% tool-call correctness**.

**Software Development Intern** — Feb. 2024 – May 2024
*Defence Science and Technology Agency* — *Singapore, SG*

- Built a **real-time 2D rapid-mapping** capability that georeferences live drone video and telemetry into GeoTIFF orthoimages **during flight**; integrated into the team's Ground Control Station.
- Designed a double-buffered, four-layer **GeoServer** rendering architecture that raised capacity from **70 to 1,100 frames** (approx. 37 min of flight), served as standard OGC WMS layers.
- Developed a post-flight telemetry review application (**React**, **PostgreSQL/PostGIS**) showing drone position, imagery and computed camera field of view on a time slider.

**Senior Geospatial Mapper (National Service)** — Jan. 2022 – Jan. 2024
*Digital and Intelligence Service, Singapore Armed Forces* — *Singapore, SG*

- Led approximately **100 personnel** at the Singapore Armed Forces Mapping Unit, Singapore's national mapping agency, overseeing operations and training.
- Produced and validated geospatial products in **QGIS** and **ArcGIS**, integrating heterogeneous sources across coordinate reference systems and map projections for military and government clients under time-sensitive requirements.

## Leadership

**President** Feb. 2026 – Present
*Manchester Artificial Intelligence Society* *Manchester, UK*

- Lead the official AI/Data Science society at the University of Manchester with over 2,300 members; increased active member count by **220 members**.
- **Organised 20+ events** (technical workshops, hackathons, outreach events) in collaboration with partners such as **BNY, UiPath, JetBrains**.
- Delivered **introductory lectures on the fundamental concepts of Machine Learning** to 60 new members.
- Led a collaboration with **University of Manchester IT Services** to develop and pilot the UoM Network Agent, an AI assistant integrating university services for connectivity support.
- Established a student **Red Team** to evaluate the agent against malicious prompts and security/accuracy risks.

## Selected Technical Projects

**UoMMods – Founder & Developer** Jun. 2025 – Present

- Built and operate a **course-planning platform** for University of Manchester students, with interactive multi-year planning, prerequisite/dependency graphs, module search/filtering and course information.
- Reached approximately **1,000 unique users**, with support for **courses across 5 departments.**
- Developed an **agentic data-scraping pipeline** using Gemma 3 and Ollama to extract structured programme information from heterogeneous university webpages.
- Invited by the University's Deputy CIO to a **strategic planning workshop with IT Services and Teaching & Learning leadership** on data-driven student systems.

**Poker-Bot Competition Platform** Nov. 2026

- Co-built the platform for a poker-bot competition run as part of **GreatUniHack Manchester**, drawing **73 teams** and **247 participants**.
- Extended the evaluation backend with **game-state logging** and **robustness and safety checks** for untrusted submissions, and added **cross-platform support** across Windows and Linux.
- Built the platform's **React** frontend, including a delayed **live-stream** of the final round for a real-time feel, and kept the frontend and backend in sync throughout.
- Designed the competition format (two rounds, fixed hand counts) and served as **sole judge** on event day, handling all submissions and executing matches on an **HPC cluster**.

## Scholarships

**DSTA Overseas Scholarship** 2024

- Full-ride scholarship awarded by Singapore's **Defence Science and Technology Agency** to support undergraduate & postgraduate study in STEM courses.

## Awards

**Best Overall Performance – 1st Prize**, University of Manchester 2026

- Awarded for achieving the **highest overall academic performance** in second year among a cohort of approximately 250 students.

**Netcraft Prize**, University of Manchester 2025

- Awarded for achieving a position in the **top 10 students** in first year of the BSc Computer Science programme.

**BrainHack Today I Learned – AI Advanced Track Champion** Jun. 2024

- Won the **AI Advanced Track** at Singapore's largest student AI competition, competing among **1,000+ participants** in a month-long challenge spanning **speech recognition, NLP and computer vision**.

**Singapore AI Capture-the-Flag – Top 40/300 Teams** Oct. 2024

- Placed among the **top 40 of 300 teams** in Singapore's first competitive AI security challenge involving adversarial problem-solving and machine learning techniques.

## Technical Skills

**Languages**: Python, C, C++, Java, Haskell, Verilog, Assembly, JavaScript, TypeScript
**AI/ML**: PyTorch, scikit-learn, LangSmith, LangGraph, DeepEval, OR-Tools, OpenAI API (tool/function calling)
**Infrastructure / Data**: React, Next.js, PostgreSQL/PostGIS, GeoServer, OSMnx/NetworkX, Docker, Google Cloud Platform, Vercel, QGIS, ArcGIS